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GREATER NOIDA INSTITUTE OF TECHNOLOGY (Engg. Institute)

CLASS TEST-I, SESSION 2023- 2024 (ODD SEM.)

COURSE- B.Tech

DEPTT.- CSE-IoT

YEAR: - 2nd (3rd SEM)

SUBJECT NAME:- Data Structure (BCS-301)

TIME: 2.00 Hours

MAX MARKS: 30

Note: Answer all questions. Draw neat sketch, wherever necessary. Suitably assume missing data, if any.

SECTION A

Question 1: Attempt all parts in brief.

[1X5=5]

a.	Write the structure of single and double linked list.	1	[CO1], K1
b.	Mention two advantages of linked list over array.	1	[CO1], K2
c.	How Sparse matrix is different from normal matrix?	1	[CO1], K1
d.	What is worst case analysis?	1	[CO1], K1
e.	Mention two differences between stack and queue.	1	[CO2], K2

SECTION B

Question 2: Attempt any 05 out of the following.

[3X5=15]

a.	Given an array arr[1.....10][1.....15] with a base value of 100 and the size of each element is 1 Byte in memory. Find the address of arr[8][6] with the help of row major order.	3	[CO1], K2
b.	Explain the representation of multidimensional arrays.	3	[CO1], K1
c.	Write a C program to print upper triangle matrix.	3	[CO1], K2
d.	What do you mean by asymptotic notation?	3	[CO1], K1
e.	Write the algorithm for tower of Hanoi problem.	3	[CO2], K2
f.	A circular queue circular queue has a size of 5 and has 3 elements 10,20 and 40 where F=2 and R=4. After inserting 50 and 60, what is the value of F and R. Trying to insert 30 at this stage what happens? Delete 2 elements from the queue. Show the sequence of steps with necessary diagrams with the value of F & R.	3	[CO2], K2
g.	Explain priority queue.	3	[CO2], K2

SECTION C

Question 3: Attempt any 02 out of the following.

[5X2=10]

a.	Evaluate the following postfix expression using stack. $2\ 3\ 9\ * +\ 2\ 3\ ^ -\ 6\ 2\ / +$, show the contents of each and every steps, also find the equivalent prefix form of above expression. Where ^ is an exponent operator.	5	[CO1], K2
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b.	What is circular linked list? Write the algorithm to insert element at the end and beginning of a circular linked list.	5	[CO1], K2
c.	Suppose multidimensional arrays P and Q are declared as P (-2:2, 2:22) and Q(1:8, -5:5, -10:5) stored in column major order. (i) Find the length of each dimension of P and Q. (ii) The number of elements in P and Q. (iii) Assuming Base Address (Q) = 400, W = 4. Find the effective address of the element Q[3,3,3].	5	[CO2], K2

Bloom's Taxonomy					
K1	K2	K3	K4	K5	K6
Remember	Understand	Apply	Analyze	Evaluate	Create